

Work with Flow to Debug Your Program

Flow is a static type checker for JavaScript. It enables programmers to write clean code by quickly finding errors in JavaScript applications.



There are two types of programming languages—strongly typed and weakly typed. Strongly typed languages are those in which the various constructs of the programming language, which include variables, expressions and functions, have a particular type. This type cannot be altered. For example, you cannot concatenate a string with an integer because these are of two different types. On the other hand, weakly typed programming languages are flexible with the type their constructs can hold, and the type of construct they actually hold. Here, concatenation of strings and numbers is possible, though these are of different types.

For example, consider JavaScript and Python. In JavaScript, we can add a string to a number like this...

```
String s= "You have a meeting at" + 10 + "AM tomorrow";  
Output : You have a meeting at 10 AM tomorrow
```

...whereas, in the case of Python, adding a number to

a string will result in an error because here, this kind of interaction between two types (string and int) is not allowed. So, we have to convert the number '10' to a string first, and then add it to the string. This is because JavaScript is weakly typed and Python is strongly typed. We can see that the type system affects how we interact with the programming language, and how we code.

Hence, we can say that every programming language has a type system. When we try to run a code snippet that violates its type rule, the programming language throws a type error.

The type system is used to reduce the chances of error by checking if the parts have been connected in a consistent way — for example, checking whether a value that is transported to another location (via a function call, an assignment, etc) has the correct type.

Static type checking performs the type checking operation before the execution of the program. To perform this